E1 244 Detection and Estimation Theory
Assignment 6

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Problem 1

Ans:

(a)

Here

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, \quad \mathbf{y} - \mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \quad \mathbf{y} - \mathbf{x}_1 \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}).$$

$$f(\mathbf{y} \mid H_0) = k \exp\left[-\frac{1}{2}\left(\frac{\|\mathbf{y} - \mathbf{x}_0\|_2^2}{\sigma^2}\right)\right], \quad \mathbf{x}_0 = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix},$$

$$f(\mathbf{y} \mid H_1) = k \exp\left[-\frac{1}{2}\left(\frac{\|\mathbf{y} - \mathbf{x}_1\|_2^2}{\sigma^2}\right)\right], \quad \mathbf{x}_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix},$$

where k is constant.

The binary hypothesis testing problem becomes

$$\frac{f(\mathbf{y} \mid H_1)}{f(\mathbf{y} \mid H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma.$$

$$L(\mathbf{y}) = \exp\left[\frac{1}{2\sigma^2}\left(\|\mathbf{y} - \mathbf{x}_0\|_2^2 - \|\mathbf{y} - \mathbf{x}_1\|_2^2\right)\right].$$

$$\Rightarrow \quad \frac{1}{2\sigma^2}\left(\|\mathbf{y} - \mathbf{x}_0\|_2^2 - \|\mathbf{y} - \mathbf{x}_1\|_2^2\right) \underset{H_0}{\overset{H_1}{\gtrless}} \ln \gamma.$$

$$\Rightarrow \quad 2\mathbf{y}^T(\mathbf{x}_1 - \mathbf{x}_0) + \|\mathbf{x}_0\|_2^2 - \|\mathbf{x}_1\|_2^2 \underset{H_0}{\overset{H_1}{\gtrless}} 2\sigma^2 \ln \gamma.$$

$$\Rightarrow \quad \mathbf{y}^T(\mathbf{x}_1 - \mathbf{x}_0) \underset{H_0}{\overset{H_1}{\gtrless}} \sigma^2 \ln\gamma - \frac{1}{2}\left[\|\mathbf{x}_0\|_2^2 + \|\mathbf{x}_1\|_2^2\right].$$

Sufficient statistic

$$T = \mathbf{y}^T(\mathbf{x}_1 - \mathbf{x}_0) = \mathbf{d}^T\mathbf{y}, \quad \mathbf{d} = \mathbf{x}_1 - \mathbf{x}_0.$$

(b)

$$T = \mathbf{d}^T\mathbf{y}$$
$$T - \mathbf{d}^T\mathbf{x}_0 \sim \mathcal{N}(0, \|\mathbf{d}\|_2^2\sigma^2), \quad T - \mathbf{d}^T\mathbf{x}_1 \sim \mathcal{N}(0, \|\mathbf{d}\|_2^2\sigma^2).$$

$$P_F = \Pr(T > \eta \mid H_0) = \int_\eta^\infty \frac{1}{\sigma\|\mathbf{d}\|_2\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{T - \mathbf{d}^T\mathbf{x}_0}{\sigma\|\mathbf{d}\|_2}\right)^2\right] dT$$
$$= Q\left(\frac{\eta - \mathbf{d}^T\mathbf{x}_0}{\sigma\|\mathbf{d}\|_2}\right).$$

Similarly,

$$P_D = Q\left(\frac{\eta - \mathbf{d}^T\mathbf{x}_1}{\sigma\|\mathbf{d}\|_2}\right).$$

$$Q^{-1}(P_F) = \frac{\eta - \mathbf{d}^T\mathbf{x}_0}{\sigma\|\mathbf{d}\|_2}$$

$$\Rightarrow \quad P_D = Q\left(Q^{-1}(P_F) + \frac{\mathbf{d}^T\mathbf{x}_0}{\sigma\|\mathbf{d}\|_2} - \frac{\mathbf{d}^T\mathbf{x}_1}{\sigma\|\mathbf{d}\|_2}\right) = Q\left(Q^{-1}(P_F) - k\right),$$

where

$$k = \frac{\mathbf{d}^T(\mathbf{x}_1 - \mathbf{x}_0)}{\sigma\|\mathbf{x}_1 - \mathbf{x}_0\|_2} = \frac{\|\mathbf{x}_1 - \mathbf{x}_0\|_2}{\sigma}.$$

(c)

From part (b),

$$P_D = g(P_F) = Q\left(Q^{-1}(P_F) - k\right).$$

This has the same form as the ROC of the DC signal in noise problem, with

$$d = \frac{\|\mathbf{x}_1 - \mathbf{x}_0\|_2}{\sigma}.$$

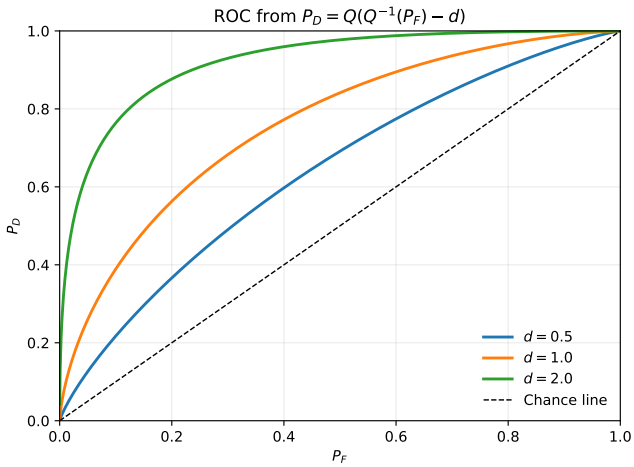

Figure 1: ROC curves from $P_D = Q(Q^{-1}(P_F) - d)$ for different d.

Problem 2

Ans:

(a)

Total area $= c \times 12$.

For pdf,

$$12c = 1 \quad \Rightarrow \quad c = \frac{1}{12}.$$

$$P(H_0) = P(H_1) = \frac{1}{2}.$$

(b)

$$f(y, x \leq 0) = f(y, H_0) = f(y \mid H_0)P(H_0) \Rightarrow f(y \mid H_0) = 2f(y, H_0).$$

For $x \leq 0$,

$$\begin{aligned} 0 \leq y \leq 2 \quad &\text{with value } \frac{1}{12} \times 2 = \frac{1}{6}, \\ -2 \leq y < 0 \quad &\text{with value } \frac{1}{12} \times 2 = \frac{1}{6}, \\ 1 \leq y \leq 2 \quad &\text{with value } \frac{1}{12} \times 2 = \frac{1}{6}, \\ -2 \leq y \leq -1 \quad &\text{with value } \frac{1}{12} \times 2 = \frac{1}{6}. \end{aligned}$$

Hence,

$$f(y \mid H_0) = \begin{cases} \frac{1}{3}, & -2 \le y \le -1 \text{ or } 1 \le y \le 2, \\ \frac{1}{6}, & -1 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

For $x > 0$,

$$\begin{aligned} 0 \le y \le 2 \quad & \text{with value } \frac{1}{12} \times 2 = \frac{1}{6}, \\ -2 \le y \le 0 \quad & \text{with value } \frac{1}{12} \times 2 = \frac{1}{6}, \\ 0 \le y \le 1 \quad & \text{with value } \frac{1}{12} \times 2 = \frac{1}{6}, \\ -1 \le y \le 0 \quad & \text{with value } \frac{1}{12} \times 2 = \frac{1}{6}. \end{aligned}$$

Hence,

$$f(y \mid H_1) = \begin{cases} \frac{1}{6}, & -2 \le y \le -1 \text{ or } 1 \le y \le 2, \\ \frac{1}{3}, & -1 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

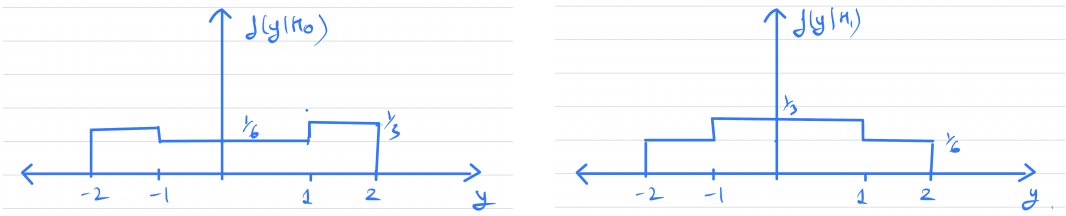

Figure 2: Sketches of $f(y \mid H_0)$ and $f(y \mid H_1)$ for Problem 2(b).

(c)

$$L(y) = \frac{f(y \mid H_1)}{f(y \mid H_0)}.$$

$$-1 \le y \le 1: \quad L(y) = \frac{1/3}{1/6} = 2,$$

$$1 \le y \le 2 \text{ and } -2 \le y \le -1: \quad L(y) = \frac{1/6}{1/3} = \frac{1}{2}.$$

For uniform cost, decision rule

$$L(y) \underset{H_0}{\overset{H_1}{\gtrless}} 1.$$

So,

$$\delta(y) = \begin{cases} H_1, & -1 \le y \le 1, \\ H_0, & 1 \le y \le 2 \text{ and } -2 \le y \le -1. \end{cases}$$

(d)

$$P(\text{error}) = P(H_1 \mid H_0)P(H_0) + P(H_0 \mid H_1)P(H_1)$$
$$= \frac{1}{2}\int_{-1}^{1} f(y \mid H_0)\,dy + \frac{1}{2}\cdot 2\int_{1}^{2} f(y \mid H_1)\,dy \quad [f(y \mid H_1) \text{ is even}]$$
$$= \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.$$

Problem 3

Ans:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{bmatrix}, \quad H_j,\ j = 0, 1, \ldots, M-1, \quad p_{ij} = \Pr\{y_i = 1 \mid H_j\}.$$

(a)

$$f(\mathbf{y} \mid H_j) = \prod_{i=1}^{d} f(y_i \mid H_j) = \prod_{i=1}^{d} p_{ij}^{y_i}(1-p_{ij})^{1-y_i} \quad [\text{independent}].$$

Define

$$f_j(\mathbf{y}) = \ln\big(f(\mathbf{y} \mid H_j)P(H_j)\big) = \sum_{i=1}^{d} y_i \ln\frac{p_{ij}}{1-p_{ij}} + \sum_{i=1}^{d}\ln(1-p_{ij}) + \ln P(H_j).$$

So,

$$f(\mathbf{y} \mid H_j)P(H_j) \propto e^{f_j(\mathbf{y})}.$$

$$\begin{aligned}\Pr(\text{Error}) = \Pr(\delta(\mathbf{y}) \neq H) &= \sum_{j=0}^{M-1} \sum_{i=0,\, i\neq j}^{M-1} \Pr(\hat{H} = H_i, H = H_j) \\ &= \sum_{j=0}^{M-1} \Big(1 - \Pr(\hat{H} = H_j, H = H_j)\Big).\end{aligned}$$

In order to minimize Pr(Error), we need to maximize

$$\sum_{j=0}^{M-1} \Pr(\hat{H} = H_j, H = H_j) = \max \sum_{j=0}^{M-1} \int_{Z_j} f(\mathbf{y} \mid H_j)P(H_j)\, d\mathbf{y}$$

$$= \max \left(\int_{Z_0} e^{f_0(\mathbf{y})} d\mathbf{y} + \int_{Z_1} e^{f_1(\mathbf{y})} d\mathbf{y} + \cdots + \int_{Z_{M-1}} e^{f_{M-1}(\mathbf{y})} d\mathbf{y} \right).$$

Choose regions:

$$\begin{aligned} Z_0 &= \{\mathbf{y} : f_0(\mathbf{y}) \geq f_i(\mathbf{y})\ \forall i \neq 0\}, \\ Z_1 &= \{\mathbf{y} : f_1(\mathbf{y}) \geq f_i(\mathbf{y})\ \forall i \neq 1\}, \\ &\vdots \\ Z_{M-1} &= \{\mathbf{y} : f_{M-1}(\mathbf{y}) \geq f_i(\mathbf{y})\ \forall i \neq M-1\}. \end{aligned}$$

So, assign H_j as

$$j = \arg\max_j f_j(\mathbf{y}).$$

That is the decision rule given in the question.

(b)

If $M = 2$, d be odd, $P(H_0) = P(H_1) = 0.5$.

Decision rule reduces to

$$\delta(\mathbf{y}) = \begin{cases} H_1, & f_1(\mathbf{y}) > f_0(\mathbf{y}), \\ H_0, & f_0(\mathbf{y}) > f_1(\mathbf{y}). \end{cases}$$

Now $f_1(\mathbf{y}) > f_0(\mathbf{y})$ gives

$$\begin{aligned}\sum_{i=1}^{d} y_i \ln \frac{p_{i1}}{1-p_{i1}} + \sum_{i=1}^{d} \ln(1-p_{i1}) + \ln P(H_1) > \sum_{i=1}^{d} y_i \ln \frac{p_{i0}}{1-p_{i0}} \\ + \sum_{i=1}^{d} \ln(1-p_{i0}) + \ln P(H_0).\end{aligned}$$

With $p_{i0} = p$, $p_{i1} = 1 - p$,

$$\sum_{i=1}^{d} y_i \left[\ln \frac{1-p}{p} - \ln \frac{p}{1-p} \right] + \sum_{i=1}^{d} [\ln p - \ln(1-p)] > 0.$$

$$\Rightarrow \quad 2 \ln \frac{1-p}{p} \sum_{i=1}^{d} y_i > d \ln \frac{1-p}{p}.$$

Since $p < \frac{1}{2}$,

$$\sum_{i=1}^{d} y_i > \frac{d}{2}.$$

So H_1 is decided when $\sum_{i=1}^{d} y_i > \frac{d}{2}$, H_0 otherwise.

(c)

$$P_e = P(H_0 \mid H_1)P(H_1) + P(H_1 \mid H_0)P(H_0)$$
$$= \sum_{\sum y_i \le d/2} \Pr(\mathbf{y} \mid H_1)P(H_1) + \sum_{\sum y_i > d/2} \Pr(\mathbf{y} \mid H_0)P(H_0).$$

$$\Pr(\mathbf{y} \mid H_0) = \prod_{i=1}^{d} \Pr(y_i \mid H_0) = \prod_{i=1}^{d} p^{y_i}(1-p)^{1-y_i} = p^{\sum y_i}(1-p)^{d-\sum y_i}.$$

Summing Bernoulli independent trials is Binomial distribution.
So second term becomes

$$\sum_{k=d/2+1}^{d} \binom{d}{k} p^k (1-p)^{d-k}.$$

First term is

$$\sum_{k=0}^{d/2} \binom{d}{k} (1-p)^k p^{d-k}.$$

Both are basically the same term, so

$$P(\text{error}) = \frac{1}{2} \sum_{k=0}^{d/2} \binom{d}{k} (1-p)^k p^{d-k} + \frac{1}{2} \sum_{k=0}^{d/2} \binom{d}{k} (1-p)^k p^{d-k}$$

$$= \sum_{k=0}^{d/2} \binom{d}{k} (1-p)^k p^{d-k}.$$

Problem 4

Ans:

$$H_k: \; Y \sim \text{Poisson}((k+1)\lambda), \quad k = 0, 1, 2, \ldots, M-1.$$

(a)

Optimal Bayes test for M-ary with uniform cost and equal priors is ML detector.

$$\hat{k} = \arg \max_{k \in \{0,1,\ldots,M-1\}} f(y \mid H_k).$$

Here,

$$f(y \mid H_i) = \frac{e^{-(i+1)\lambda}((i+1)\lambda)^y}{y!}.$$

So,

$$\delta(y) = \begin{cases} H_0, & \text{if } f(y \mid H_0) \geq f(y \mid H_i) \; \forall i \neq 0, \\ H_1, & \text{if } f(y \mid H_1) \geq f(y \mid H_i) \; \forall i \neq 1, \\ \vdots & \\ H_{M-1}, & \text{if } f(y \mid H_{M-1}) \geq f(y \mid H_i) \; \forall i \neq M-1. \end{cases}$$

(b)

Pick up a decision region say H_j.

Then,

$$f(y \mid H_j) \geq f(y \mid H_i), \quad \forall i \neq j.$$

$$\Rightarrow \quad \frac{e^{-(j+1)\lambda}(j+1)^y \lambda^y}{y!} \geq \frac{e^{-(i+1)\lambda}(i+1)^y \lambda^y}{y!}, \quad \forall i \neq j$$

$$\Rightarrow \quad e^{-j\lambda}(j+1)^y \geq e^{-i\lambda}(i+1)^y$$

$$\Rightarrow \quad -j\lambda + y\ln(j+1) \geq -i\lambda + y\ln(i+1)$$

$$\Rightarrow \quad y \ln \frac{j+1}{i+1} \geq (j-i)\lambda$$

$$\Rightarrow \quad y \geq \frac{(j-i)\lambda}{\ln \frac{j+1}{i+1}}.$$

So, to decide H_k

$$\frac{\lambda}{\ln \frac{k+2}{k+1}} \geq y \geq \frac{\lambda}{\ln \frac{k+1}{k}}.$$

(c)

$$P(\text{error}) = \sum_{i=0}^{M-1} \Pr(\text{error} \mid H_i) P_i = \sum_{i=0}^{M-1} \left[1 - \Pr(H_i \mid H_i)\right] P_i.$$

Now,

$$\Pr(H_i \mid H_i) = \sum_{y=\left\lceil \frac{\lambda}{\ln \frac{i+2}{i+1}} \right\rceil}^{\left\lfloor \frac{\lambda}{\ln \frac{i+1}{i}} \right\rfloor} \frac{e^{-(i+1)\lambda}((i+1)\lambda)^y}{y!}.$$

$$P_i = \frac{1}{M}.$$

Problem 5

Ans:

(a)

For $y < 0$,

$$P_F = \int_{-\infty}^{0} \frac{1}{2} f(y \mid H_0)\, dy = \frac{1}{2} \int_{-\infty}^{0} \frac{1}{2} e^{-|y|}\, dy = \frac{1}{4} \int_{-\infty}^{0} e^{y}\, dy = \frac{1}{4}.$$

For $0 \leq y \leq \tau$,

$$P_F = \int_{0}^{\tau} \frac{1}{2} e^{-|y|}\, dy = \frac{1}{2}(1 - e^{-\tau}).$$

For $y > \tau$, $P_F = 0$.

For $y < 0$,

$$P_D = \frac{1}{2} \int_{-\infty}^{0} f(y \mid H_1)\, dy = \frac{1}{2} \int_{-\infty}^{0} e^{-2|y|}\, dy = \frac{1}{2} \int_{-\infty}^{0} e^{2y}\, dy = \frac{1}{4}.$$

For $0 \leq y \leq \tau$,

$$P_D = \int_{0}^{\tau} f(y \mid H_1)\, dy = \int_{0}^{\tau} e^{-2|y|}\, dy = \int_{0}^{\tau} e^{-2y}\, dy = \frac{1 - e^{-2\tau}}{2}.$$

So,

$$P_F = \frac{1}{4} + \frac{1}{2}(1 - e^{-\tau}), \qquad P_D = \frac{1}{4} + \frac{1}{2}(1 - e^{-2\tau}).$$

$$1 - 2\left(P_F - \frac{1}{4}\right) = e^{-\tau}.$$

$$\Rightarrow \quad P_D = \frac{1}{4} + \frac{1}{2}\left[1 - \left\{1 - 2\left(P_F - \frac{1}{4}\right)\right\}^2\right]$$

$$= \frac{1}{4} + 2\left(P_F - \frac{1}{4}\right) - \left(P_F - \frac{1}{4}\right)^2.$$

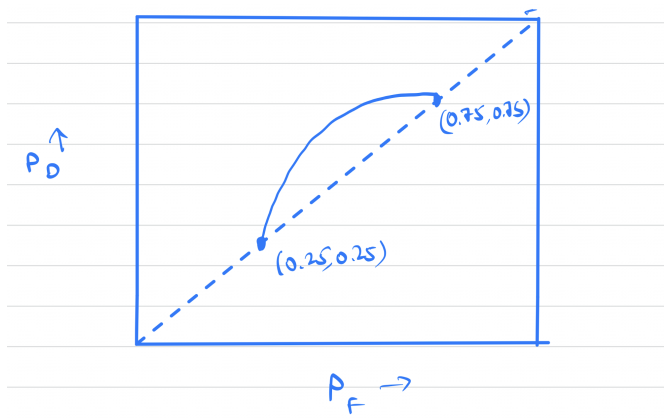

Figure 3: ROC sketch for Problem 5(a).

(b)

Let a randomization be defined between $(0, 0)$ and $(\alpha, g(\alpha))$, where $(\alpha, g(\alpha))$ lies on ROC curve of $\delta_3(y)$.

Slope of the curve:

$$\frac{dP_D}{dP_F} = \frac{dP_D}{d\tau} \cdot \frac{d\tau}{dP_F} = \frac{e^{-2\tau}}{\frac{1}{2}e^{-\tau}} = 2e^{-\tau}.$$

Slope of the line from origin:

$$\frac{P_D}{P_F} = \frac{\frac{3}{4} - \frac{1}{2}e^{-2\tau}}{\frac{3}{4} - \frac{1}{2}e^{-\tau}}.$$

For tangency,

$$\frac{P_D}{P_F} = \frac{dP_D}{dP_F} \Rightarrow \frac{\frac{3}{4} - \frac{1}{2}e^{-2\tau}}{\frac{3}{4} - \frac{1}{2}e^{-\tau}} = 2e^{-\tau}.$$

$$\Rightarrow \quad \frac{3}{4} - \frac{1}{2}e^{-2\tau} = \frac{3}{2}e^{-\tau} - e^{-2\tau}$$

$$\Rightarrow \quad -\frac{1}{2}e^{-2\tau} - \frac{3}{2}e^{-\tau} + \frac{3}{4} = 0$$

$$\Rightarrow \quad e^{-2\tau} + 3e^{-\tau} - 1.5 = 0$$

$$\Rightarrow \quad e^{-\tau} = \frac{1}{\sqrt{2}}.$$

$$\Rightarrow \quad P_F = \frac{3}{4} - \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}.$$

This is the required $\alpha_0 = \dfrac{\sqrt{2}}{4}$.
Similarly, by symmetry,

$$\alpha_1 = 1 - \alpha_0.$$

(c)
We know from part (b),

$$P_F \in [0, \alpha_0] \;\rightarrow\; \text{straight line (from randomization)}$$

$$P_F \in [\alpha_0, \alpha_1] \;\rightarrow\; \text{curve of } \delta_3$$

$$P_F \in [\alpha_1, 1] \;\rightarrow\; \text{straight line (from randomization).}$$

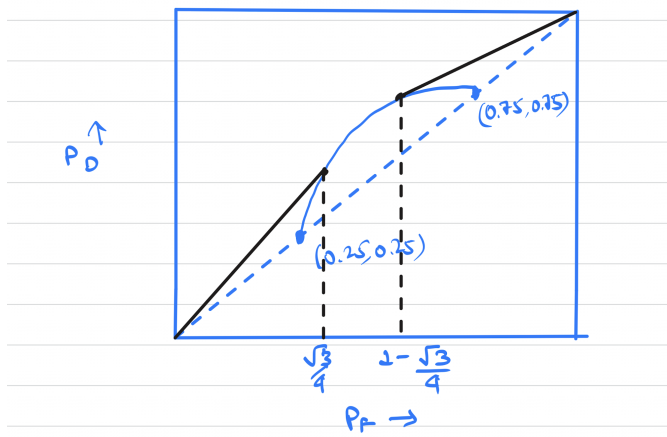

Figure 4: ROC sketch for Problem 5(c).